Maxime Munari

PhD candidate • ENS de Lyon • Machine Learning & Signal Processing

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PROFILE

PhD candidate in Physics at ENS de Lyon working on machine learning, statistical learning, and signal processing for high-dimensional scientific data, with a focus on robust learning, leakage-free evaluation, and weak-signal analysis.

RESEARCH EXPERIENCE

Sept. 2025 – Present — **Doctoral Researcher** — *Supervisor: Pierre Borgnat*
ENS de Lyon — Laboratoire de Physique
Development of machine learning and signal processing methods for high-dimensional mass spectrometry data in biomedical applications, focusing on weak-signal detection and robust statistical learning.

Mar. 2025 – Jul. 2025 — **Master's Thesis** — *Supervisor: Denis Bartolo*
ENS de Lyon — Laboratoire de Physique
Computer vision and machine learning for collective behaviour analysis and tracking of fish schools in active matter experiments.

Feb. 2024 – May. 2024 — **Research Internship** — *Supervisor: Gourab Saha*
IPHC Strasbourg / CERN (CMS)
Graph Neural Networks in PyTorch for quark/gluon jet discrimination on $p-p$ collision simulations.

Mar. 2025 – Jul. 2025 — **Research Internship** — *Supervisor: Sylvain Torre*
Laboratoire d'Astrophysique de Marseille
Analysis of galaxy distributions in simulated Euclid catalogues. Redshift measurement and error propagation.

EDUCATION

2024 — 2025 **MSc, Numerical Physics** • CY Paris University
2023 — 2024 **MSc, Physics (1st year)** • Strasbourg University
2020 — 2023 **BSc, Physics** • Grenoble University

TECHNICAL SKILLS

Programming	Python • LaTeX • Git/GitHub • Java • C++
Machine Learning	PyTorch • PyTorch Geometric • scikit-learn
Scientific Computing	NumPy • SciPy • pandas • matplotlib • h5py
Tools & Environment	macOS • Linux • Jupyter • HPC • multiprocessing